

Cisco Networking & Cybersecurity Internship Report

Comprehensive Virtual Summer Internship Report on Network Simulation, Architecture Protocols, and Cybersecurity Essentials

Abhinav Yadav

Student Trainee | Cisco Networking Academy Program

Summer Internship 2026

IILM University Academy Track

About The Student

CANDIDATE PROFILE



Abhinav Yadav

Cisco Certified Student Intern

INSTITUTION

IILM University, Greater Noida

ACADEMIC TRACK

B.Tech CSE

INTERNSHIP DOMAIN

Networking & Security

PROGRAM DURATION

June – August 2026

Internship Summary

- ✓ Completed intensive technical pathways in Cisco Networking and Cybersecurity Fundamentals.
- ✓ Mastered network configuration, topology modeling, and routing principles using Cisco Packet Tracer.
- ✓ Gained practical knowledge in identifying cyber threats, risk mitigation, and security policies.
- ✓ Earned 4 industry-recognized Cisco Networking Academy global certification credentials.

About The Program

ECOSYSTEM PARTNERS



Cisco Networking Academy

Global IT and cybersecurity education program delivering industry-aligned curriculum, hands-on simulation tools, and real-world network engineering skillsets.



IILM University Academy

Institutional host facilitating structured instructor-led virtual learning modules, lab practice, and continuous assessment mentorship under certified faculties.



National Skill Standard

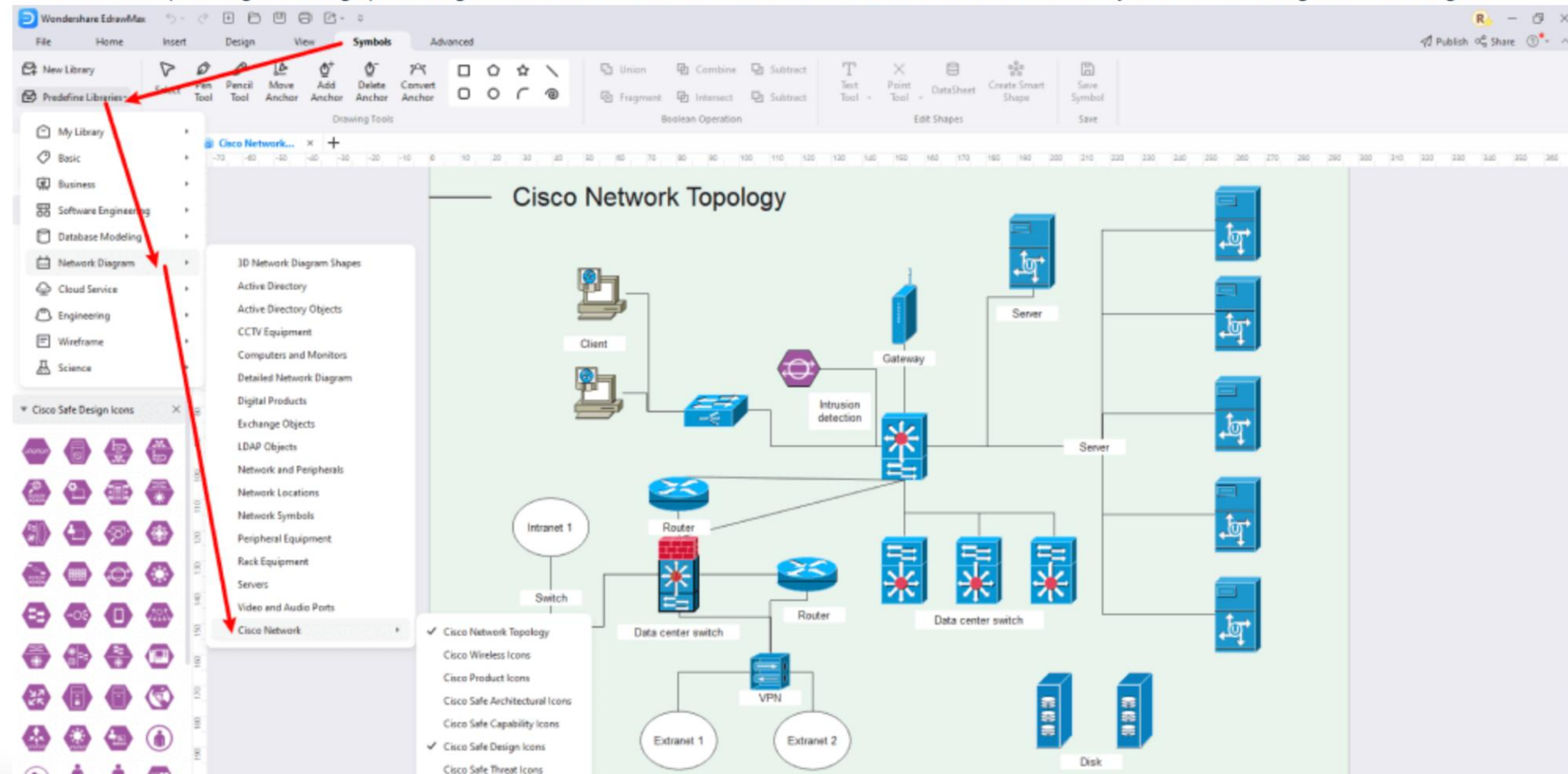
Designed to equip engineering students with foundational knowledge in computer networks, threat detection, data privacy, and infrastructure security.

Problem Statement

Bridging Theory & Practical Network Security

While theoretical computer networking and cybersecurity are core computer science subjects, students often lack direct hands-on practice in configuring real-world network topologies and analyzing live security threats.

This internship bridges this gap through interactive Packet Tracer simulation and real-world cyber threat mitigation strategies.



Project Objectives

KEY GOALS



Packet Tracer Mastery

Design, configure, and troubleshoot complex LAN/WAN network topologies, routers, switches, and wireless access points using Cisco Packet Tracer.



Cybersecurity Foundations

Understand threat actors, malware vectors, encryption standards, and defense strategies to protect organizational digital assets.

Technologies & Tools

TECHNICAL STACK



Cisco Packet Tracer

Network simulation & visualization
software



Network Topologies

LAN, WAN, Switches & Router
Configuration



IP Addressing

IPv4, IPv6, Subnetting & DHCP
Protocols



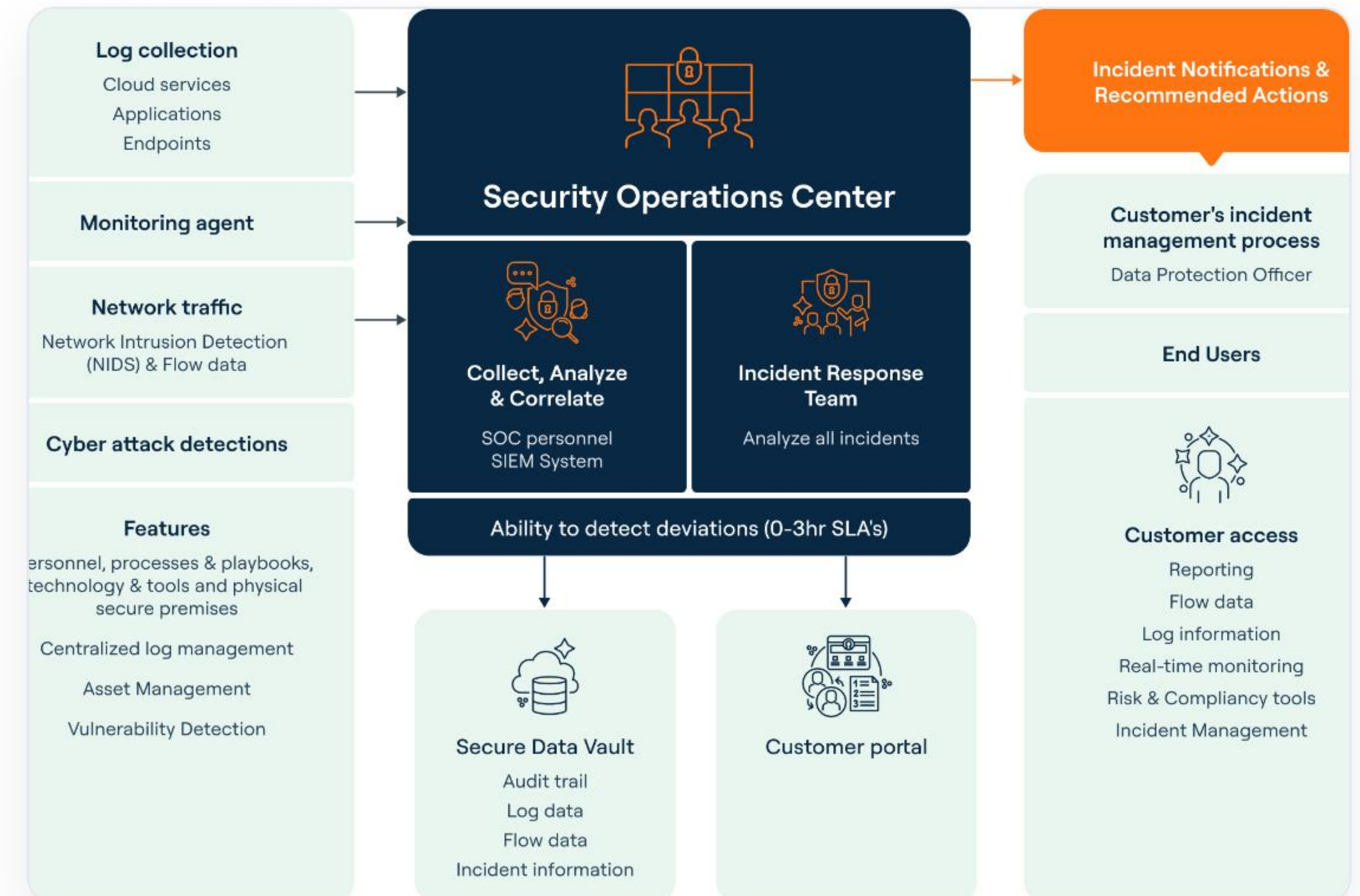
Threat Defense

Firewalls, Vulnerability & Malware
Assessment

Learning Architecture

CURRICULUM ROADMAP

- ▶ **Module 1: Getting Started with Packet Tracer**
Basic GUI overview, device connections, and media types.
- 🔧 **Module 2: Exploring Networking with Packet Tracer**
Simulating routing protocols, switches, and IoT connectivity.
- 🔒 **Module 3: Cybersecurity Essentials**
Deep dive into confidentiality, integrity, availability (CIA triad).
- 👤 **Module 4: Introduction to Cybersecurity**
Global threat landscape, cyber defense laws, and ethics.



Learning Journey

EXECUTION TIMELINE

1

13 JUNE 2026

Packet Tracer Exploration

Explored networking simulation & connected nodes.

2

16 JUNE 2026

Packet Tracer Basics

Mastered network creation & terminal CLI.

3

20 JUNE 2026

Security Essentials

Studied threat actors, firewalls & security principles.

4

29 AUG 2026

Intro to Cybersecurity

Final credential in cybersecurity concepts.

Key Learning Outcomes

COMPETENCIES GAINED

- ✓ **Network Simulation Expertise:** Ability to build, model, and troubleshoot network topologies using Cisco Packet Tracer software.
- ✓ **Protocol Understanding:** Practical knowledge of TCP/IP stack, Ethernet, wireless protocols, and IP addressing schemes.
- ✓ **Cyber Threat Identification:** Knowledge of common security vulnerabilities including phishing, ransomware, and social engineering.
- ✓ **Data Protection Standards:** Clear understanding of privacy protocols, encryption algorithms, and security compliance frameworks.

Completed Certifications

VERIFIED CREDENTIALS

CISCO NETWORKING ACADEMY

Introduction to Cybersecurity

Awarded by Cisco Networking Academy Director Lynn Bloomer.

Date: 29 Aug 2026

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Cybersecurity Essentials

Instructor: Swati Vashisht | IILM University

Date: 20 Jun 2026

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IILM UNIVERSITY / CISCO

Exploring Networking with Cisco Packet Tracer

Instructor: Mr Abhinav Raghav | IILM University

Date: 13 Jun 2026

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IILM UNIVERSITY / CISCO

Getting Started with Cisco Packet Tracer

Instructor: Mr Abhinav Raghav | IILM University

Date: 16 Jun 2026

791f99a5-4bdb-4af5-825c-954dc5887893

Thank You!

Questions & Answers

Abhinav Yadav



IILM University, Greater Noida

Image Sources



<https://images.edrawsoft.com/articles/cisco-network-diagram/step4.png>

Source: www.edrawsoft.com



<https://netnordic.com/app/uploads/2024/10/Illustration-Security-Operations-Center.png>

Source: netnordic.com